CSE322: Introduction to Embedded Systems

Fall 2026

Project Description

Home Appliances Control

Each team of **4-6 members** builds a Bluetooth-controlled home-appliance prototype on the TM4C123GH6PM board, with a mobile app, and submits code, report and demo video by Week 13.

Project Overview

The prototype lets a phone control and monitor a small home through the TM4C123GH6PM board, over Bluetooth. The mobile app and the board must do the following:

- Switch a 220 V lamp on and off from the app. Wire the relay contacts in parallel with the existing manual switch, so the manual switch keeps working.
- Switch a power plug on and off from the app. When the plug relay is off, the plug has no power.
- Show the door status (open or closed), read from a magnetic reed switch.
- Show the room temperature in the app, refreshed every second, read from an LM35 sensor.
- Let the user set a temperature limit in the app. Above the limit, the app shows an alert.
- Save the time of every door opening and closing in the app, and show the list.

Hardware Requirements

Each team needs these components:

- One TM4C123GH6PM LaunchPad board
- One HC-05 Bluetooth module
- One LM35 temperature sensor
- One magnetic reed switch and its magnet, for the door (or a pushbutton)
- One 5 V relay module with 2 channels, one for the lamp and one for the plug
- One external 5 V power supply (at least 1 A) for the relays, LM35 and HC-05
- A 220 V lamp with its manual switch, and a power plug
- A panel to mount everything on, with the wires hidden
- A mobile phone for the app

Pin map

Every team uses the same pins, so one test setup can check all teams. Power the relays, LM35 and HC-05 from one external 5 V power supply. Connect the supply's GND to a GND pin of the LaunchPad so that all grounds are common. The LaunchPad itself is powered by its USB cable.

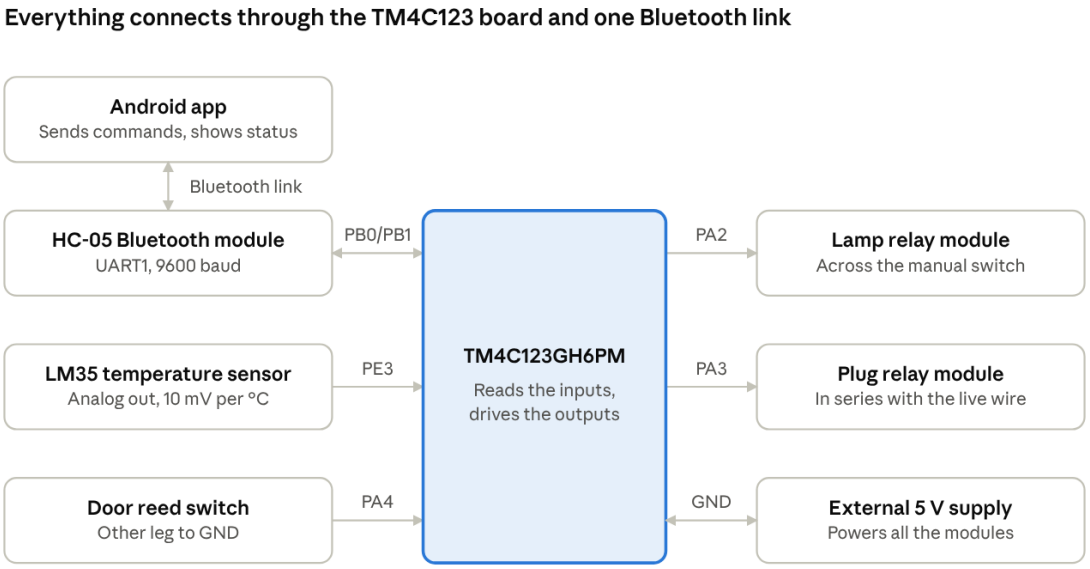
Component	Module pin	TM4C123 pin
Lamp relay module	IN	PA2
Plug relay module	IN	PA3
Door reed switch	one leg (other leg to GND)	PA4
LM35 temperature sensor	OUT	PE3 (AIN0)
HC-05 Bluetooth module	TXD	PB0 (U1Rx)
HC-05 Bluetooth module	RXD	PB1 (U1Tx)

Bluetooth Communication

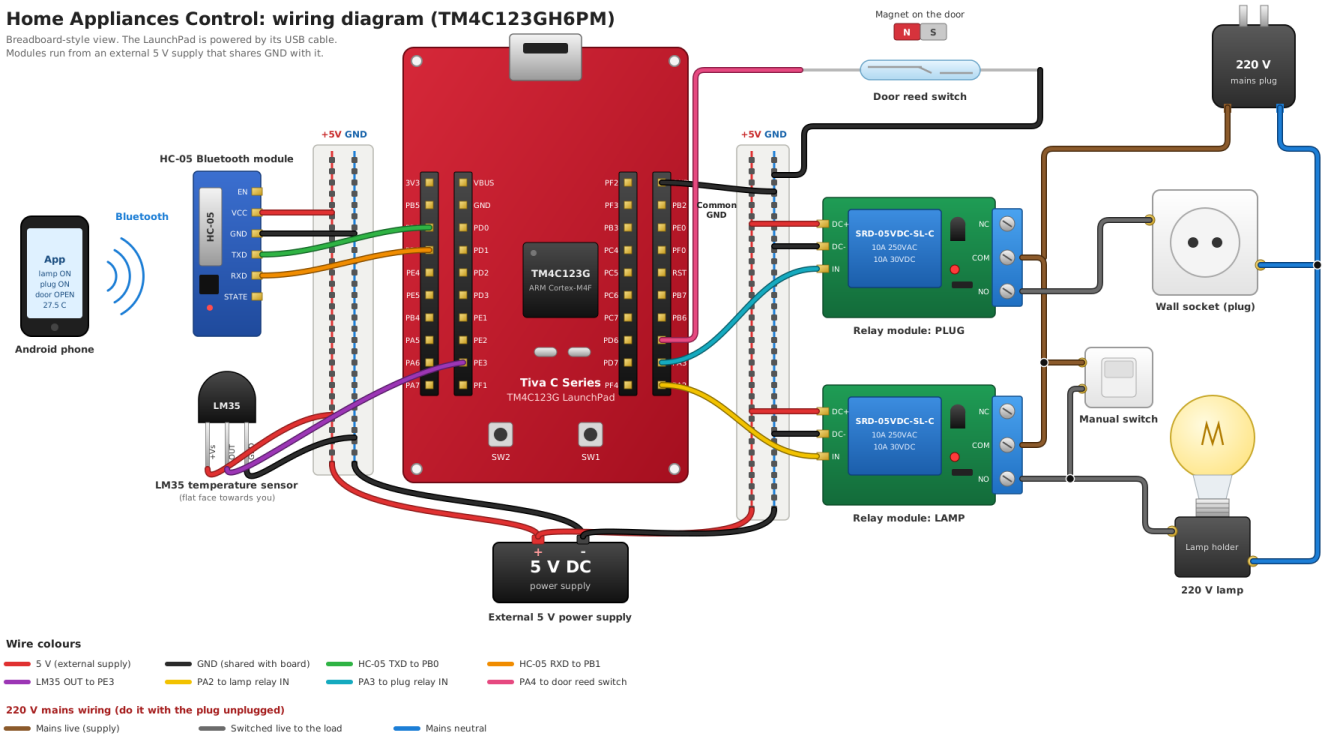
The app and the board exchange ASCII text lines, each ending in a newline, at 9600 baud. The app stamps each door message with the phone's clock when it saves the door log.

Direction	Message	Meaning
App to board	LAMP:1 or LAMP:0	Lamp relay on or off
App to board	PLUG:1 or PLUG:0	Plug relay on or off
App to board	LIMIT:30	Set the temperature limit to 30 °C (any whole number)
App to board	STATUS?	Board replies with lamp, plug, door, temperature and alarm state
Board to app	TEMP:27.5	Room temperature in °C, sent every second
Board to app	DOOR:OPEN or DOOR:CLOSED	Sent on every door change
Board to app	ALARM:1 or ALARM:0	1 when temperature is above the limit, 0 when it returns to the limit or below
Board to app	ACK:LAMP:1	Confirms a command; repeats the command received

Connection Diagram



Wiring Diagram



Project Deliverables

Each team submits one compressed file by the end of Week 13, containing:

1. The project folder: the TM4C123 firmware source (following the pin map and protocol above), the mobile app project.
2. A demo video of 5 minutes showing every requirement working on the real prototype.
3. A technical report in one PDF file with:
    - The contribution of each team member.
    - Flowcharts of the main program flow.
    - The list of components and the wiring diagrams.
    - Photos of the prototype.
    - Problems faced and how the team solved them.

Grading Rubric

Criteria	Marks	Details
1) Functional Requirements	**17**	
– Lamp control	3	App on/off through the relay, manual switch still works, state shown in the app
– Plug control	2	App on/off; the plug has no power when off
– Door status	2	Reed switch read on the board, open/closed shown live in the app
– Temperature display	2	LM35 read through the ADC, value refreshed every second in the app
– Temperature alarm	2	Limit set in the app; alert shown when the temperature is above it
– Door event log	2	Time of every opening and closing saved and listed in the app
– Bluetooth communication	2	Reliable commands and replies as in the protocol table
– Mounting and wiring	2	Components mounted on a suitable panel, wires hidden
3) Software Design & Report	**3**	
– Layered Architecture	1	Clear separation into MCAL, HAL and Application layers
– Final Report	2	Well-structured and complete with all required sections
Total	**20**	